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PRAYCG Control Center v0.5 Installation Guide

Document version: v1.0

Target audience: Windows users who are only mildly familiar with Python and software installation.

Software target: PRAYCG_ControlCenter_v0_5_SelfContained_AllSoftware.zip

1. What this guide installs

This guide installs the software needed to run the **PRAYCG Control Center v0.5** and the bundled PRAYCG toolchain.

The v0.5 bundle contains the Control Center plus the current PRAYCG software tree:

```
PRAYCG_ControlCenter_v0_5_SelfContained/  
  control_center/  
  tools/  
    MediaPrep_StimulusFingerprint_v1_8/  
    ProtocolRunner_PRAYCG2_0/  
    MasterComprehensiveSuite_v1_5_6_CURRENT/  
    Offline_Master_Interpreter_v1_5_6/  
    Acquisition/  
      BrainFlow_EEGOnly_Home_Bridge_v1_0/  
      BrainFlow_ALS_PT19_Bridge_v1_4/  
      Polar_H10_to_LSL/  
      Vernier_Respiration_to_LSL/  
  docs/
```

The bundle does **not** include these external applications:

```
Python 3.11  
PsychoPy / PsychoPy Coder  
LabRecorder  
OpenBCI drivers / board drivers
```

The Control Center has a **Use Bundled Paths** button. That points its settings at the software already inside the v0.5 folder. It also has **Auto-find ALL**, which searches for external tools such as PsychoPy and LabRecorder.

2. Important boundaries

PRAYCG is exploratory research software. It is not a medical device, diagnostic tool, therapy tool, or proof of consciousness, memory biology, OSM biology, or any metaphysical mechanism.

For actual human recordings, follow consent, privacy, and ethics requirements appropriate to your setting. Do not upload private biosignal recordings or unredacted self-report to a public repository. Do not redistribute copyrighted films or clips.

3. Recommended Windows folder layout

Use short paths. This avoids Windows path-length errors.

Recommended:

```
C:\PRAYCG_CC_v05\  
C:\PRAYCG\  
C:\LabRecorder\
```

Avoid long paths like:

```
C:\Users\YourName\Downloads\Some Very Long Folder Name\Another Folder\Another  
Folder\PRAYCG...
```

The install steps below assume you will extract the PRAYCG bundle to:

```
C:\PRAYCG_CC_v05\
```

4. How to open Command Prompt

1. Press the **Windows** key.
2. Type:
cmd

3. Press **Enter**.

A black terminal window should open. This is **Command Prompt**.

To paste into Command Prompt, use **Ctrl+V** or right-click inside the window.

You usually do **not** need Administrator mode for Python user installs. Use Administrator only if Windows asks for it or if you intentionally install for all users.

5. Check whether winget is available

In Command Prompt, run:

```
winget --version
```

If it prints a version number, continue.

If Windows says winget is not recognized, install or update **App Installer** from the Microsoft Store. Microsoft documents WinGet as the Windows Package Manager command-line tool available on Windows 10/11 through App Installer. See the reference links at the end of this document.

You can also try:

```
powershell Add-AppxPackage -RegisterByFamilyName -MainPackage  
Microsoft.DesktopAppInstaller_8wekyb3d8bbwe
```

Then close Command Prompt, reopen it, and run:

```
winget --version
```

again.

6. Install Python 3.11 with winget

Run:

```
winget source update
```

Search for Python 3.11:

```
winget search --id Python.Python.3.11
```

Install Python 3.11:

```
winget install -e --id Python.Python.3.11 --accept-source-agreements --accept-package-  
agreements
```

If you want to force a user-level install, try:

```
winget install -e --id Python.Python.3.11 --scope user --accept-source-agreements --accept-  
package-agreements
```

If --scope user fails, rerun the first command without --scope user.

After installation, close Command Prompt and open a new Command Prompt.

Verify Python:

```
py -3.11 --version
```

Expected result:

Python 3.11.x

Also check pip:

```
py -3.11 -m pip --version
```

If py -3.11 works, use it for all PRAYCG commands.

7. Download and extract PRAYCG Control Center v0.5

1. Download:

PRAYCG_ControlCenter_v0_5_SelfContained_AllSoftware.zip

2. Create a short folder:

```
mkdir C:\PRAYCG_CC_v05
```

3. Extract the ZIP into:

C:\PRAYCG_CC_v05\

After extraction, you should have:

C:\PRAYCG_CC_v05\PRAYCG_ControlCenter_v0_5_SelfContained\

Open that folder in File Explorer and confirm these files exist:

```
run_PRAYCG_ControlCenter_v0_5.bat
install_common_requirements_windows.bat
requirements_PRAYCG_v0_5_common.txt
README_START_HERE.md
VERSION_MANIFEST_v0_5.json
control_center\scripts\praycg_control_center_v0_5.py
```

8. Install Python requirements - easiest method

Open Command Prompt.

Go to the PRAYCG v0.5 folder:

```
cd /d C:\PRAYCG_CC_v05\PRAYCG_ControlCenter_v0_5_SelfContained
```

Upgrade pip:

```
py -3.11 -m pip install --upgrade pip setuptools wheel
```

Install the common PRAYCG requirements:

```
py -3.11 -m pip install -r requirements_PRAYCG_v0_5_common.txt
```

Install the Control Center optional requirements:

```
py -3.11 -m pip install -r control_center\requirements_optional.txt
```

If the command fails while installing psychopy, do not panic. The recommended runner workflow uses **PsychoPy Standalone / PsychoPy Coder**, not the Python 3.11 environment. Continue with the rest of the install, then install PsychoPy Standalone separately.

9. Install Python requirements - selective method

If the full requirements command is slow or fails, install the important packages in smaller groups.

Core analysis / media packages:

```
py -3.11 -m pip install numpy scipy pandas matplotlib opencv-python moviepy pillow  
pyxdf pylsl mne pyyaml openpyxl
```

BrainFlow/OpenBCI acquisition packages:

```
py -3.11 -m pip install brainflow pyserial pylsl numpy
```

Polar H10 BLE package:

```
py -3.11 -m pip install bleak pylsl
```

Vernier Go Direct Respiration Belt package:

```
py -3.11 -m pip install godirect pylsl numpy
```

Offline interpreter packages:

```
py -3.11 -m pip install numpy pandas
```

Then run:

```
py -3.11 -m pip check
```

If pip check reports warnings, read them. Warnings do not always prevent the Control Center from opening, but they may affect specific modules.

10. Install PsychoPy Standalone

The PRAYCG2.0 runner is a PsychoPy experiment. On the current working system, the runner has been reliable when launched from **PsychoPy Coder**.

Recommended install:

1. Go to the official PsychoPy download page.
2. Download the current **PsychoPy Standalone** installer for Windows.
3. Run the installer.
4. Open PsychoPy once.
5. Open **Coder** from PsychoPy.
6. Close PsychoPy.

Do not rely on direct Python launch if the runner does not work that way on your machine. The Control Center supports the safer workflow:

Open PsychoPy Coder + runner file/folder

If PsychoPy reports No module named pylsl when running the PRAYCG runner, install or expose pylsl inside the PsychoPy environment. PsychoPy has a package/plugin workflow, but the exact interface can vary by PsychoPy version. A practical workaround is to use PsychoPy's package manager or pip terminal, if available, and install:

pylsl

If that fails, document the error and use the Control Center's logs to diagnose the missing package.

11. Install LabRecorder

PRAYCG records LSL streams into .xdf files using **LabRecorder**.

Recommended install:

1. Go to the LabRecorder GitHub releases page.
2. Download the Windows release archive.
3. Extract it to:

C:\LabRecorder\

4. Confirm this file exists:

C:\LabRecorder\LabRecorder.exe

If the program does not open, install the Microsoft Visual C++ Redistributable if Windows asks for it.

LabRecorder should show available LSL streams after you click **Update**. It records selected streams into one .xdf file.

12. Install or test OpenBCI board support

Before using PRAYCG, verify that Windows can see the OpenBCI dongle/board.

Recommended:

1. Install and open the official OpenBCI GUI.
2. Confirm the Cyton + Daisy can stream there first.
3. Note the COM port used by the board, for example:

COM3

COM4

COM5

4. Close OpenBCI GUI before launching BrainFlow. Do not let two programs fight for the same serial port.

If Windows does not recognize the dongle, install the appropriate USB/serial driver for your OpenBCI hardware.

13. First launch of the Control Center

Open Command Prompt.

Go to the bundle folder:

```
cd /d C:\PRAYCG_CC_v05\PRAYCG_ControlCenter_v0_5_SelfContained
```

Start the Control Center:

```
run_PRAYCG_ControlCenter_v0_5.bat
```

Alternative direct launch:

```
py -3.11 control_center\scripts\praycg_control_center_v0_5.py
```

If the Control Center opens, go to the **Settings** tab.

Click:

Use Bundled Paths

This points the Control Center to the bundled PRAYCG tools.

Then click:

Auto-find ALL

This searches for external applications such as PsychoPy and LabRecorder.

Review the settings before launching anything.

14. What Use Bundled Paths should populate

The **Use Bundled Paths** button should point to these internal tools:

MediaPrep GUI:

tools\MediaPrep_StimulusFingerprint_v1_8\scripts\praycg_media_prep_gui_v1_8.py

PRAYCG2.0 Runner:

tools\ProtocolRunner_PRAYCG2_0\scripts\run_PRAYCG2_0_ConsolidatedSelfReport.py

Master Comprehensive Suite:

tools\MasterComprehensiveSuite_v1_5_6_CURRENT\...

Offline Interpreter:

tools\Offline_Master_Interpreter_v1_5_6\scripts\praycg_offline_interpretive_report_gui_v1_5_6.py

BrainFlow EEG-only:

tools\Acquisition\BrainFlow_EEGOnly_Home_Bridge_v1_0\scripts\praycg_openbci_brainflow_eeg_to_lsl_v1_0.py

BrainFlow EEG+ALS:

tools\Acquisition\BrainFlow_ALS_PT19_Bridge_v1_4\scripts\praycg_openbci_brainflow_als_pt19_bridge_v1_4.py

Polar H10:

tools\Acquisition\Polar_H10_to_LSL\polar_to_lsl.py

Vernier:

tools\Acquisition\Vernier_Respiration_to_LSL\vernier_respiration_belt_to_lsl.py

15. Start the Control Center from File Explorer

You can also double-click:

run_PRAYCG_ControlCenter_v0_5.bat

If a black window appears and quickly closes, open Command Prompt manually and run the .bat from there so you can read the error.

16. Test the Control Center installation

Inside the Control Center:

1. Open **Settings**.
2. Click **Use Bundled Paths**.
3. Click **Auto-find ALL**.
4. Open **Acquisition**.
5. Click **Check Visible LSL Streams**.

It is okay if no streams appear yet. That only means no streamers are running.

Then test a simple bundled script path:

1. Open **Analysis**.
2. Try launching the **Offline Interpreter** GUI.
3. Close it.

If that works, Python and the Control Center are functioning.

17. Test Polar H10 LSL bridge

1. Wear the Polar H10 strap.
2. Wet the electrodes.
3. Make sure Bluetooth is on.
4. In the Control Center, go to **Acquisition**.
5. Click **Start Polar H10 RR -> LSL**.
6. Watch the log window.

Expected behavior:

Scanning for Polar H10...

Found Polar H10...

LSL Stream 'PolarHRV' established.

Beat Caught: ### ms

The stream name is:

PolarHRV

This is an RR-interval stream. It is not a continuous clinical ECG waveform.

18. Test Vernier Respiration Belt LSL bridge

USB is recommended first.

1. Plug the Vernier Go Direct Respiration Belt into USB.
2. Put the belt around your torso.
3. In the Control Center, go to **Acquisition**.
4. Click **Start Vernier Respiration USB -> LSL**.
5. Watch the log.

Expected stream name:

VernierRespirationBelt

The default channel is:

force_N

Use force_N for offline inhale/exhale phase reconstruction.

19. Test BrainFlow EEG-only

1. Plug in the OpenBCI dongle.
2. Turn on the Cyton + Daisy board.
3. Find the COM port. You can use the included identify-port script or the Windows Device Manager.
4. In Control Center, open **Acquisition**.
5. Use **Start BrainFlow EEG Only**.

If the script needs a COM port and the GUI does not know it yet, open the BrainFlow tool folder and run the example batch files:

tools\Acquisition\BrainFlow_EEGOnly_Home_Bridge_v1_0\examples\

Edit the COM port if needed.

Expected EEG stream name is usually:

obci_eeg1

20. Test BrainFlow EEG + ALS/PT19

Use this only after EEG-only works.

1. Plug in the ALS/PT19 light sensor according to your BrainFlow bridge wiring.
2. Place the ALS/PT19 over the timing square or use the Control Center ALS barcode test.
3. Open **Acquisition**.
4. Click **Start BrainFlow EEG + ALS**.
5. Click **Check Visible LSL Streams**.

Expected streams may include:

obci_eeg1

OpenBCIStatusMarkers

Analog Read / AUX channel stream, depending on bridge version

For timing validation, the ALS signal should rise sharply during the white pulse and return to baseline after the pulse.

21. Test LabRecorder

1. Start at least one LSL stream, such as Polar or BrainFlow.
2. Open LabRecorder.
3. Click **Update**.
4. Confirm streams appear.
5. Check the boxes for streams you want to record.
6. Set the save folder.
7. Click **Start**.
8. Wait a few seconds.
9. Click **Stop**.
10. Confirm an .xdf file was written.

A PRAYCG recording should eventually include streams such as:

obci_eeg1

OpenBCIStatusMarkers

PolarHRV

VernierRespirationBelt

praycg_stasismarkers_...

22. Recommended install verification checklist

Before a real run, confirm:

- [] Python 3.11 works with `py -3.11 --version`.
- [] PRAYCG Control Center opens.
- [] Settings -> Use Bundled Paths was clicked.
- [] Settings -> Auto-find ALL found PsychoPy and/or LabRecorder.
- [] PsychoPy Coder opens.
- [] LabRecorder opens.
- [] Polar H10 stream appears as PolarHRV.
- [] Vernier stream appears as VernierRespirationBelt.
- [] BrainFlow EEG-only stream appears as obci_eeg1.

- [] BrainFlow EEG+ALS stream works if using ALS/PT19.
 - [] LabRecorder can record a short test XDF.
-

23. Common errors and fixes

winget is not recognized

Install or update **App Installer** from the Microsoft Store. Reopen Command Prompt.

py is not recognized

Close and reopen Command Prompt. If it still fails, reinstall Python 3.11 and ensure the Python Launcher is installed.

No module named pylsl

Run:

```
py -3.11 -m pip install pylsl
```

If this happens inside PsychoPy Coder, install pylsl into PsychoPy's environment or use PsychoPy's package manager/pip terminal if available.

No module named cv2

Run:

```
py -3.11 -m pip install opencv-python
```

No module named brainflow

Run:

```
py -3.11 -m pip install brainflow
```

Polar H10 not found

Check:

- Bluetooth on
- strap wet and worn
- strap not already connected to another app
- battery good
- near the computer

Vernier device not found

Use USB first. Close other Vernier apps. Try a different USB port.

BrainFlow cannot open COM port

Close OpenBCI GUI or any other program using the board. Check the COM port in Device Manager.

LabRecorder shows no streams

Start streamers first. Click **Update** in LabRecorder. Check firewall and local network settings if streams are on another computer.

PRAYCG runner does not work from direct Python launch

Use the recommended workflow:

Control Center -> Runner/PsychoPy -> Open PsychoPy Coder + runner file/folder

Then press Run from inside PsychoPy Coder.

24. Reference links

- Microsoft WinGet overview: <https://learn.microsoft.com/windows/package-manager/winget/>
 - Microsoft WinGet install command: <https://learn.microsoft.com/windows/package-manager/winget/install>
 - PsychoPy download page: <https://psychopy.org/download.html>
 - PsychoPy external modules: <https://psychopy.org/tutorials/addCustomModules.html>
 - LabRecorder GitHub: <https://github.com/labstreaminglayer/App-LabRecorder>
 - Lab Streaming Layer: <https://labstreaminglayer.org/>
-

25. Final rule

If you only remember one install rule, remember this:

Install Python 3.11.

Extract PRAYCG v0.5 to a short path.

Install requirements.

Open Control Center.

Click Use Bundled Paths.

Click Auto-find ALL.

Launch the PRAYCG2.0 runner through PsychoPy Coder.